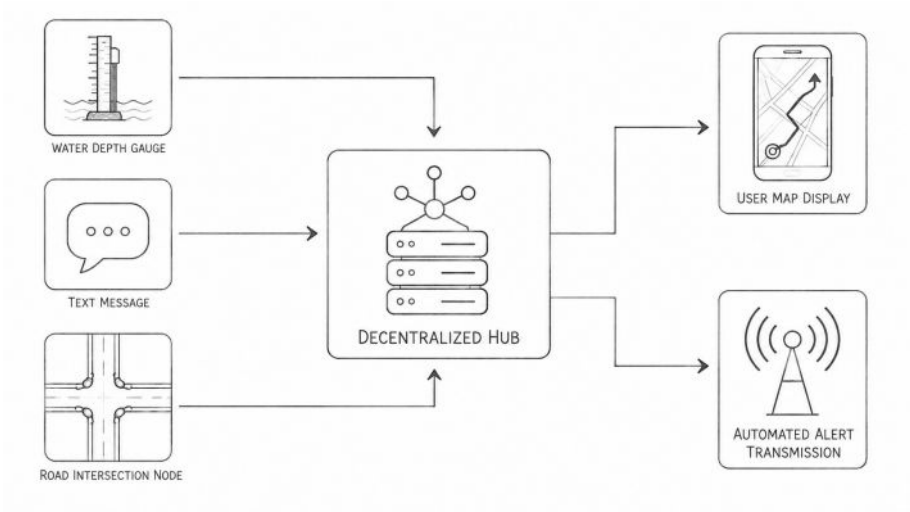
Campaign
IEEE Response Quest Challenge



Tabaport: On-Site Edge Intelligence & Dispatch System for Urban Flash Floods (#5267)

Author:

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05/24/2026 - 6:15am | Impact Challenge Product Submission | Confidential

Is this a Team Submission?

No

Disaster Scenario

Severe torrential downpours in Kampala, Uganda, trigger rapid urban flash floods that turn major transport corridors (e.g., Clock Tower Junction, Bwaise, Kalerwe) into impassable hazards within minutes. This recurring crisis paralyzes traffic, drowns commercial centers, and claims lives. The situation deteriorates rapidly because the city's drainage network (such as the Nakivubo channel) is chronically bottlenecked by massive plastic waste accumulation and heavy siltation.

During these severe storms, centralized cellular networks frequently experience total outages or extreme bandwidth congestion. This leaves municipal field engineers (KCCA), emergency dispatchers, and traffic police blind on the ground. First responders lack immediate visibility to determine which specific drainage conduits are overflowing, which emergency routes are safely passable, and where to deploy personnel to clear debris or block streets before vehicles are swept away. This proposal addresses the acute lack of resilient, on-site, real-time situational awareness when traditional communication infrastructure fails.

Concept Overview

System Summary:
TabaPort Kampala is a decentralized, offline-first edge intelligence and emergency routing system designed to guide on-site responders and protect the public during urban flash floods. The platform removes reliance on continuous cloud connectivity by handling data collection, noise filtering, priority assessment, and spatial routing entirely at the field infrastructure boundary.

Core Components and Operations:

1. **The Edge Sensor Network (Data Collection):**
   Low-cost, solar-powered IoT telemetry nodes are anchored inside concrete drainage culverts at known flooding hotspots. Each node features an ESP32 microcontroller paired with an IP67 waterproof ultrasonic depth sensor. The node continuously measures water level heights. Instead of transmitting raw, volatile data, the local firmware runs a moving median algorithm that filters out signal spikes caused by passing solid plastic trash or floating debris, ensuring clean hydrological metrics.
2. **Decentralized Edge Hubs (Data Fusion & Prioritization):**
   Data from the culvert nodes, alongside crowdsourced field status reports parsed from a backup SMS/USSD gateway, is transmitted via local radio links (LoRa/BLE) or cellular networks to a localized Edge Hub (a rugged field laptop or Raspberry Pi deployed with first responders). The hub hosts a standalone SpatiaLite database containing static OpenStreetMap (OSM) layers of Kampala’s road and drainage networks.
3. **The Analytical Hierarchy Process (AHP) Dispatch Engine:**
   To prevent data overload for stressed field crews, the Edge Hub runs a localized AHP matrix. It automatically scores and ranks flooding locations based on three dynamic weights: velocity of water rise, proximity to critical infrastructure (hospitals, major flyovers), and neighborhood population density. This transforms chaotic raw numbers into a clear, prioritized checklist of intervention sites.
4. **Offline Dynamic Navigation & Routing:**
   Field responders access a lightweight mobile dashboard. When an asset or a sensor flags a road segment as "submerged," the local mapping engine applies an infinite weight penalty to that specific road segment ID. Using a localized A* search algorithm, the app instantly recalculates safe alternative navigation paths over high-ground routes (e.g., the Northern Bypass) for all field vehicles, keeping emergency crews moving safely around hazards.
5. **KCCA Automated Public Information Loop:**
   To securely close the loop between infrastructure and the public, the system triggers an automated public notification framework:

- *Hyper-Local SMS Broadcasts:* If a node updates to critical ("Red"), the system immediately prompts the SMS gateway to issue zero-cost, location-targeted cell text alerts to all active mobile devices within a 2km radius of the hazard via MTN/Airtel networks (e.g., "Clock Tower submerged, divert immediately").
- *Automated Social Media Amplification:* The edge hub transmits compressed traffic maps via Webhooks directly to KCCA's official social feeds (X/Twitter, Facebook) using

targeted hashtags like #KampalaTraffic to instantly inform the general public and commuter networks (Uber/SafeBoda).

Why It Fits the IEEE Response Quest:
This concept perfectly targets the challenge's core objective: enabling near-real-time situational awareness for on-site personnel. It demonstrates how multi-modal data fusion (IoT, USSD text, and GIS vectors) can be processed entirely at the edge. By decoupling disaster intelligence from fragile cloud networks, *TabaPort Kampala* gives field managers the operational readiness required to make rapid, data-backed decisions that save lives and protect infrastructure.

Timeliness, Real-Time Responsiveness & Technical Reliability

The platform achieves minimal latency by executing data filtering directly on the ESP32 microcontroller. Nodes sample culvert water levels every 30 seconds, catching sudden flash spikes instantly. Processing raw distance data via an on-chip moving median filter ensures that anomalous sensor noise from floating debris is dropped immediately, eliminating upstream processing lag.

The system builds its well-structured technical foundation on a fail-safe communication hierarchy. Under standard operating conditions, nodes transmit telemetry via cellular networks. If severe storms compromise cell towers, the hardware drops back to standard GSM SMS channels to feed short, optimized data strings. If cellular coverage collapses entirely, nodes pivot to point-to-point LoRa and BLE beacon broadcasts. This allows on-site responder hubs within 500 meters to passively harvest sensor metrics.

Because the local hub maps run on an offline SpatiaLite engine, calculations for route optimizations execute instantly without cloud reliance, ensuring uninterrupted reliability.

Comprehensiveness, Use of Available Data & Novel Data Discovery

The system achieves comprehensiveness by fusing three distinct, mismatched data domains: dynamic physical IoT depth sensors, static OpenStreetMap (OSM) Uganda road and topological layers, and crowdsourced social text feeds. Novel data discovery is achieved by capturing underutilized ground-truth reports from informal transit operators (boda-boda networks) via a low-bandwidth USSD string. A regex parsing tokenizer on the edge hub cleans this text, matching location names against a geographic gazetteer to generate real-time spatial data points without internet access.

Data Limitations & Explanations:

1. *Sensor Drifts:* Heavy humidity can introduce slight ultrasonic reading drift. This is handled by running continuous baseline distance calibrations when no rain is detected.
2. *Crowdsource Anonymity:* Text inputs lack precise GPS metadata. The system flags text entries with wider structural margins of error (100-meter radius zones) on the map until an adjacent physical sensor or first responder confirms the blockage, preventing false positioning errors.

Integration, Synthesis Quality & Responsible Data Handling

Multi-modal integration is handled seamlessly by an Analytical Hierarchy Process (AHP) matrix that distills conflicting data streams into a single, highly coherent "Tactical Priority Score". By cross-referencing the physical velocity of water rise against static geographic markers (such as hospital coordinates and residential densities), the system synthesizes raw sensor data and public text alerts into a single actionable dashboard priority list for emergency managers.

Responsible Data Handling & Ethical Safeguards:
The system is built on a strict privacy-by-design approach to ensure safe data handling. The USSD/SMS ingress gateway implements an automatic scrubbing script that strips away the sender’s phone number, SIM IMSI network string, and personal identifiers at the boundary. Only the message content, timestamp, and generalized neighborhood node location are stored. Furthermore, the entire code stack is open-source (MIT licensed), keeping data collection open to public audit and preventing hidden tracking vectors.

Usability, Clarity & Operational Readiness for Emergency Responders

The application interface matches high-stress responder workflows by replacing complex charts with an intuitive, color-coded map interface. Road segments are rendered in three bold states: Green (Passable), Yellow (Rising water/Caution), and Red (Submerged/Turn back). This removes cognitive load for emergency vehicle operators during crisis scenarios.

Operational safety is maintained by an automated re-routing logic. When a road segment transitions to "Red," the A* navigation engine applies a severe travel weight penalty to that node, forcing the navigation dashboard to compute safe, high-ground alternatives (e.g., the Northern Bypass) instantly. This directly protects response crews from accidentally driving into deep valleys or open trenches. The hardware profile is completely documented in the open-source repository and relies on standard, off-the-shelf components (ESP32 microcontrollers, standard waterproof sensors). This keeps the per-node cost under $55, ensuring municipal maintenance teams can build and deploy the hardware using local tools.

Scenario Fit, Insightfulness, Innovation & Technical/Data Creativity

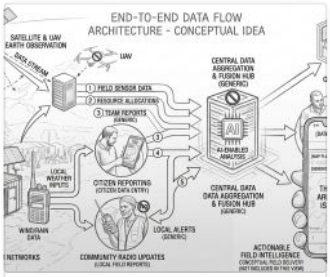
This project delivers high scenario fit by addressing the core issue of Kampala's floods: infrastructure blocking caused by plastic waste, paired with communication blackouts during storms. While standard disaster solutions rely on heavy, centralized cloud infrastructure and continuous high-bandwidth connections, *TabaPort Kampala* uses a novel, decentralized approach: it moves data processing to the edge, directly where the disaster hits.
Technical creativity is shown in its hardware and software designs. Building an affordable $5 IoT drainage gauge that clears out physical garbage noise using an on-chip moving median filter proves a deep understanding of developing urban landscapes. Fusing spatial infrastructure vectors with live crowdsourced text strings to dynamically adjust travel weights on an offline map is a highly innovative approach to emergency navigation. The solution moves past simple monitoring; it delivers resilient on-site edge intelligence that gives rescue teams the awareness needed to survive infrastructure collapses.



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How Did You Find Out About the IEEE Response Quest Challenge?

IEEE Newsletter

Information Source Details

KCCA Kampala Drainage Master Plan (kcca.go.ug); OpenStreetMap Uganda Geographic Road Network Data Layers (openstreetmap.org); Uganda National Meteorological Authority Hydro-Climate Metrics (unma.go.ug); IEEE Systems Council Resilience Guidelines.

Attestation about being the Entrant and an Active IEEE Member

I confirm that I am the Entrant and an active IEEE member.

Attestation about Original Work

I confirm that this concept is my team’s or my original work.

Acknowledgement of the IEEE Response Quest Challenge Rules

I acknowledge and agree to the IEEE Response Quest Challenge Rules

Acknowledgement of IEEE Intellectual Property Statement

I acknowledge and agree to the IEEE Intellectual Property Statement.

Concept Acceptance Comments

The submission has advanced to Phase 3 of the Challenge.

Exposure Review Team - Volunteer:

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Legal and Compliance

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JD **John D. DeSimone**
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KC **Karyn Connor**
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AN **Arash Nemati Hayati**
Information Technology

Task Management

No Results Found

Idea Adoption

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Expert Management

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Join Requests

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Additional Updates

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Exposure Review

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